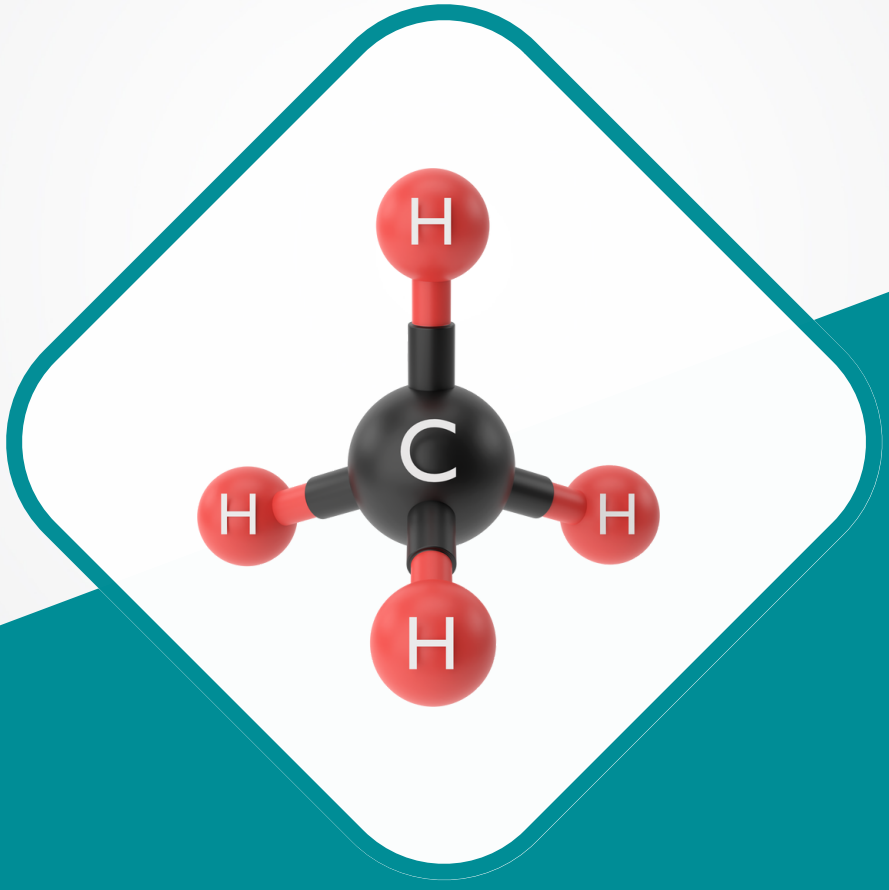
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Hydrocarbons

English Medium

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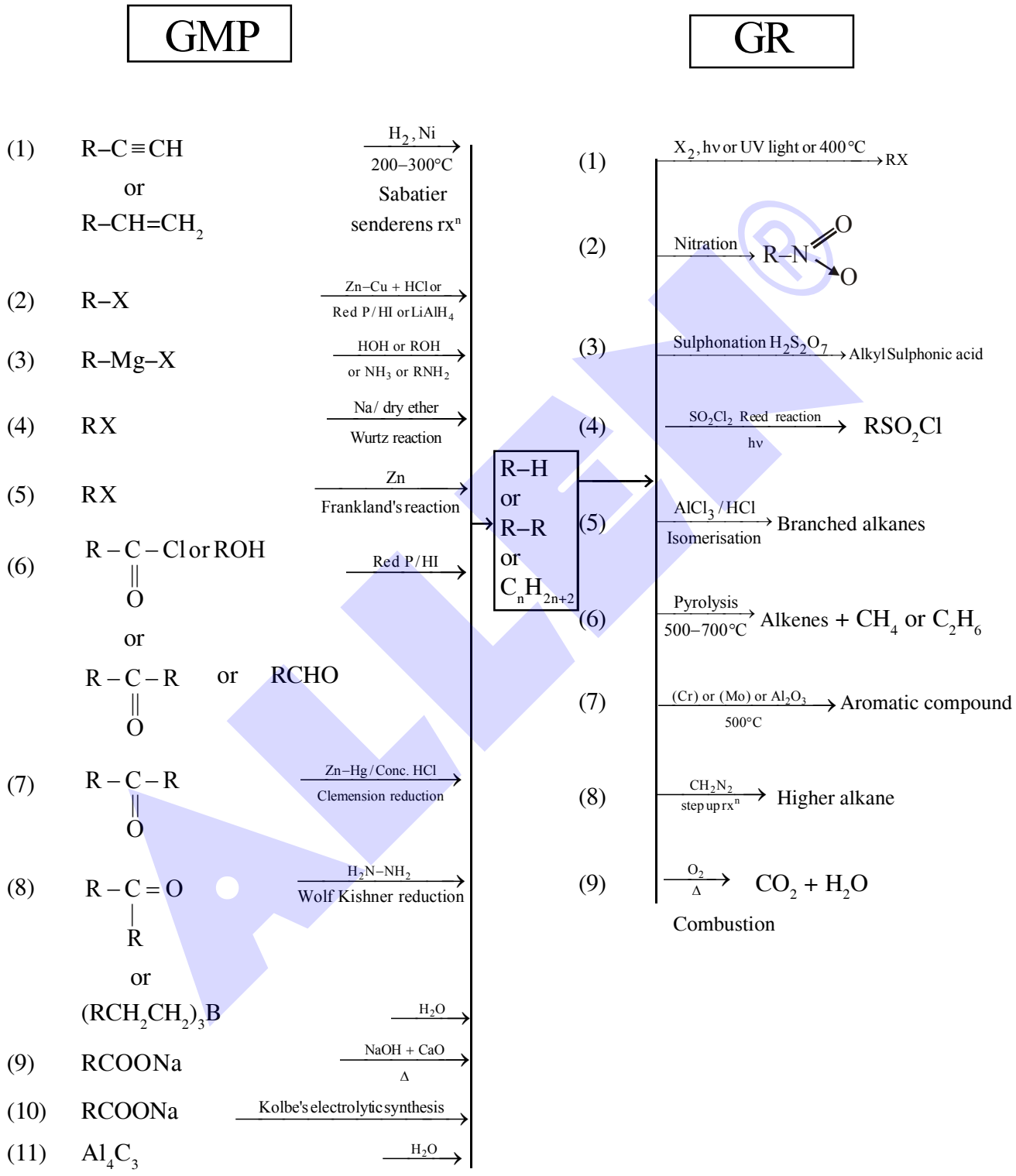
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HYDROCARBON

REACTION CHART FOR ALKANES



node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\03_Hydrocarbon\01_Theory+Ex.p65"

REACTION CHART FOR ALKENES

GMP

(1) $R–CH_2–CH_2–OH \xrightarrow[\Delta/-H_2O]{\text{conc. } H_2SO_4}$

(2) $R–CH_2–CH_2–X \xrightarrow[-HX]{\text{alc. KOH}}$

(3) $R–CH_2–CHX_2 \xrightarrow[\text{for higher alkene } -X_2]{\text{Zn dust}/\Delta}$

(4) $R-\underset{X}{CH}-\underset{X}{CH_2} \xrightarrow[\Delta]{\text{Zn dust}}$

(5) $R–C\equiv CH \xrightarrow[BaSO_4/CaCO_3]{H_2/Pd}$

(6) $\begin{matrix}RCH-COOK\\ |\\ RCH-COOK\end{matrix} \xrightarrow{\text{Kolbe's electrolytic synthesis}}$

(7) $(C_2H_5)_4N^+OH \xrightarrow{\Delta}$

(8) $R–\underset{\underset{O}{\|}}{CO}–CH_2–CH_2–R \xrightarrow{\text{Pyrolysis}}$

(9) $R–H \xrightarrow{\text{Pyrolysis}}$

(10) $CH_2=CHCl \xrightarrow[\text{(ii) Me–I}]{\text{(i) Mg}}$

(11) $R–CH=O \xrightarrow[\text{wittig reaction}]{CH_2=PPh_3}$

$R–CH=CH_2$ or C_nH_{2n}

GR

(1) $\xrightarrow[200–300°C]{H_2,\ Ni} R–CH_2–CH_3$

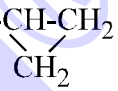
(2) $\xrightarrow{X_2} R–CHX–CH_2X$

(3) $\xrightarrow{HX} R–CHX–CH_3$

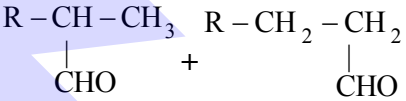
(4) $\xrightarrow{HBr,\ Peroxide} R–CH_2–CH_2Br$

(5) $\xrightarrow{HOCl} R–CH(OH)–CH_2Cl$

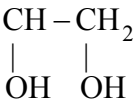
(6) $\xrightarrow[+H_2O]{\text{dil. } H_2SO_4} R–CH_2(OH)–CH_3$

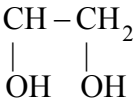
(7) $\xrightarrow{+CH_2N_2}$ 

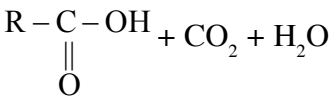
(8) $\xrightarrow{BH_3} (RCH_2CH_2)_3B$

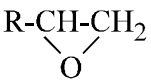
(9) $\xrightarrow[HCo(CO)_4]{CO+H_2}$ 

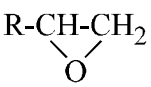
(10) $\xrightarrow[\Delta]{O_2} CO_2 + H_2O$

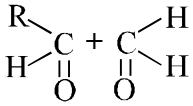
(11) $\xrightarrow[\text{(ii) } NaHSO_3/H_2O]{\text{(i) } OsO_4}$ 

(12) $\xrightarrow[1\%\text{ alkaline } KMnO_4]{\text{Baeyer's reagent}}$ 

(13) $\xrightarrow{\text{strong oxidant}}$ 

(14) $\xrightarrow[\text{Priles–chalev's reaction}]{\text{Per acid}}$ 

(15) $\xrightarrow[Ag\ 300°C]{1/2\,O_2}$ 

(16) $\xrightarrow[\text{ozonolysis}]{O_3/Zn,H_2O}$ 

(17) $\xrightarrow[200°C\text{ high P}]{O_2}$ Polyalkene

(18) $\xrightarrow[500°C]{Cl_2}$ Substitution product

(19) $\xrightarrow[200–300°C]{Al_2(SO_4)_3}$ Isomerisation

(20) $\xrightarrow[\Delta]{\text{acetic anhydride}} R–CH=CH–COCH_3$
Methyl alkenyl ketone

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket\[OC]\Eng\03_Hydrocarbon\01_Theory+Ex.p65"

REACTION CHART FOR ALKYNES

GMP

(1) $CH_2Br–CH_2Br \xrightarrow{alc.KOH,\ NaNH_2}$

(2) $CH_3–CHBr_2 \xrightarrow{alc.KOH,\ NaNH_2}$

(3) $CHCl_3 \xrightarrow[\Delta]{Ag\ powder}$

(4) $CHBr_2–CHBr_2 \xrightarrow[\Delta]{Zn\ dust}$

(5) $\begin{matrix}CHBr\\ \|\\ CHBr\end{matrix} \xrightarrow{Zn}$

(6) $CH_2=CH–Cl \xrightarrow{alc.KOH,\ NaNH_2}$

(7) $\begin{matrix}HC-COONa\\ \|\\ HC-COONa\end{matrix} \xrightarrow{Kolbe's\ electrolytic\ synthesis}$

(8) $CaC_2 \xrightarrow{H_2O}$

(9) $2C + H_2 \xrightarrow[Berthelot's\ process]{electric\ arc,1200°C}$

(10) $CH_3–C\equiv CH \xrightarrow{(i)Na\ (ii)R–X} CH_3–C\equiv CR$

(11) $CH_3–C\equiv CH \xrightarrow{(i)CH_3MgI\ (ii)R–X} CH_3–C\equiv CR$

(12) $Mg_2C_3 \xrightarrow{H_2O} CH_3–C\equiv CH$

$\rightarrow C_2H_2 \rightarrow$

GR

(1) $\xrightarrow[Ni]{H_2} C_2H_4,\ C_2H_6$

(2) $\xrightarrow{X_2} C_2H_2X_4$

(3) $\xrightarrow[Peroxide]{HBr} CH_2BrCH_2Br$

(4) $\xrightarrow[No\ peroxide]{HBr} CH_3–CHBr_2$

(5) $\xrightarrow{HOCl} Cl_2CH–COOH$

(6) $\xrightarrow{HCN,\ Ba(CN)_2} CH_2=CHCN$

(7) $\xrightarrow{CH_3COOH,\ Hg^{+2}} CH_3CH(OCOCH_3)_2$

(8) $\xrightarrow[(Kucherov's\ reaction)]{Hg^{+2},\ 80°C,\ dil.\ H_2SO_4} CH_3CHO$

(9) $\xrightarrow[Kucherov's\ reaction]{Dil.\ H_2SO_4\ /\ HgSO_4} CH_3CHO$

(10) $\xrightarrow[Ni(CO)_4]{CO+HOH} CH_2=CH–COOH$

(11) $\xrightarrow[Ni,160°C]{CO+EtOH} CH_2=CH–COOEt$

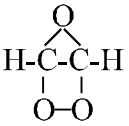
(12) $\xrightarrow{NaNH_2} NaC\equiv CNa$

(13) $\xrightarrow[(Tollen's\ Reagent)]{AgNO_3+NH_4OH} AgC\equiv CAg$ white ppt.

(14) $\xrightarrow{Cu_2Cl_2+NH_4OH} CuC\equiv CCu$ red ppt.

(15) $\xrightarrow[\Delta]{Combustion\ O_2} CO_2 + H_2O$

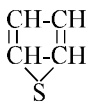
(16) $\xrightarrow{Baeyer's\ Reagent} \begin{matrix}CHO\\ |\\ CHO\end{matrix} \longrightarrow HCOOH$

(17) $\xrightarrow[Ozonolysis]{O_3}$  $\xrightarrow{+H_2O}$ 2 HCOOH

(18) $\xrightarrow[(Red\ hot\ iron\ tube)]{Trimerisation}$ benzene

(19) $\xrightarrow[[Ni(CN)_2]]{Tetramerisation}$ C_8H_8 or 1,3,5,7-cyclo octa tetraene

(20) $\xrightarrow[[Cu(NH_3)_2]^+\ or\ Cu_2Cl_2.NH_4Cl]{Dimerisation}$ butenyne

(21) $\xrightarrow[\Delta]{S}$ 

(22) $\xrightarrow[(BF_3–HgO)]{CH_3OH} \begin{matrix}CH(OCH_3)_2\\ |\\ CH_3\end{matrix}$ methylal

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydt.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\03_Hydrocarbon\01_Theory+Ex.p65"